in order for unusually high cooking temperatures to be readily achieved. The reverse is also true: When pressure is released and things are allowed to expand, they cool. If you remove the lid on a pressure cooker—or, more spectacularly, should it blow off—the air it contains will expand to its ordinary density while cooling to standard room temperature. This is the science underlying the phrase “blow off steam,” a familiar approach to “cool down” a heated situation. It turns out that these simple terrestrial observations have a profound incarnation within the cosmos.

We saw above that after electrons and nuclei join together to form atoms, photons are free to travel unimpeded through the universe, much like atoms of air in a hot but otherwise empty pressure cooker. And just as air in the pressure cooker cools down when the lid is removed and it is allowed to expand, the same is true for the “gas” of photons streaming through the universe as it expands. In fact, physicists as far back as George Gamow and his students Ralph Alpher and Robert Hermann in the 1950s, and Robert Dicke and Jim Peebles in the mid-1960s, realized that the present-day universe should be permeated by an almost uniform bath of these primordial photons, which, through the last 15 billion years of cosmic expansion, have cooled to a mere handful of degrees above absolute zero.1 In 1965, Arno Penzias and Robert Wilson of Bell Laboratories in New Jersey accidently made one of the most important discoveries of our age when they detected this afterglow of the big bang while working on an antenna intended for use with communication satellites. Subsequent research has refined both theory and experiment, culminating in measurements taken by NASA’s COBE (Cosmic Background Explorer) satellite in the early 1990s. With these data, physicists and astronomers have confirmed to high precision that the universe is filled with microwave radiation (if our eyes were sensitive to microwaves, we would see a diffuse glow in the world around us) whose temperature is about 2.7 degrees above absolute zero, exactly in keeping with the expectation of the big bang theory. In concrete terms, in every cubic meter of the universe—including the one you now occupy—there are, on average, about 400 million photons that collectively compose the vast cosmic sea of microwave radiation, an echo of creation. A percentage of the “snow” you see on your